

ANSWERS OF MODEL TEST PAPER 6

INTERMEDIATE: GROUP – II

PAPER – 6: FINANCIAL MANAGEMENT & STRATEGIC MANAGEMENT

PAPER 6A : FINANCIAL MANAGEMENT

Suggested Answers/ Hints

PART I – Case Scenario based MCQs

1. (c) ₹ 0.72

Computation of EPS under financial plan I: Equity Financing

	(₹)
EBIT	37,50,000.00
Interest	-
EBT	37,50,000.00
Less: Taxes 40%	(15,00,000.00)
PAT	22,50,000.00
No. of equity shares	31,25,000.00
EPS	0.72

2. (b) ₹ 0.90

Computation of EPS under financial plan II: Debt – Equity Mix

	(₹)
EBIT	37,50,000.00
Less: Interest	(14,06,250.00)
EBT	23,43,750.00
Less: Taxes 40%	(9,37,500.00)
PAT	14,06,250.00
No. of equity shares	15,62,500.00
EPS	0.90

3. (a) ₹ 0.44

Computation of EPS under financial plan III: Preference Shares – Equity Mix

	(₹)
EBIT	37,50,000.00
Less: Interest	-

EBT	37,50,000.00
Less: Taxes (40%)	(15,00,000.00)
PAT	22,50,000.00
Less: Pref. dividend	(15,62,500.00)
PAT for equity shareholders	6,87,500.00
No. of Equity shares	15,62,500.00
EPS	0.44

4. (a) ₹ 28,12,500

EBIT – EPS Indifference Point- Plan I and Plan II:

$$\frac{(EBIT) \times (1 - T_c)}{N_1} = \frac{(EBIT - \text{Interest}) \times (1 - T_c)}{N_2}$$

$$\frac{EBIT(1 - 0.40)}{31,25,000} = \frac{(EBIT - 14,06,250) \times (1 - 0.40)}{15,62,500}$$

$$0.6EBIT = 1.2EBIT - 16,87,500$$

$$= ₹ 28,12,500$$

5. (d) ₹ 52,08,333.33

EBIT – EPS Indifference Point: Plan I and Plan III

$$\frac{EBIT(1 - T_c)}{N_1} = \frac{EBIT(1 - T_c) - \text{Pref. Div.}}{N_2}$$

$$\frac{EBIT(1 - 0.4)}{31,25,000} = \frac{EBIT(1 - 0.4) - 15,62,500}{15,62,500}$$

$$0.6EBIT = 1.2EBIT - 31,25,000$$

$$EBIT = ₹ 52,08,333.33$$

6. (b) 30%

The formula for Degree of Combined Leverage (DCL) is:

$$DCL = DOL \times DFL$$

$$DCL = 2 \times 3 = 6$$

The percentage change in EPS is:

$$\% \Delta EPS = DCL \times \% \Delta \text{Sales}$$

$$\% \Delta EPS = 6 \times 5\% = 30\%$$

7. (c) 3.79

$$\text{Initial Investment} = \text{Annual Cost Savings} \times PVAF$$

$$\text{Annual cost savings} = ₹ 1,80,000$$

$$PVAF (10\%, 5 \text{ years}) = 3.79$$

$$\text{Initial Investment} = 1,80,000 \times 3.79 = 6,82,200$$

$$\begin{aligned}
 \text{Payback Period} &= \text{Initial Investment} / \text{Annual Cost Savings} \\
 &= 6,82,200 / 1,80,000 \\
 &= 3.79 \text{ years}
 \end{aligned}$$

8. (c) **Remains unchanged because value depends on earnings and investment policy.**

(Explanation: M&M's theory suggests that dividend policy has no impact on shareholder wealth in a perfect market.)

PART II – Descriptive Questions

1. (a) **Statement showing Computation of Combined leverage**

	₹
Sales	2,00,000
Less: Variable costs (50%)	1,00,000
Contribution	1,00,000
Less: Fixed operating costs	40,000
EBIT	60,000
Less: Interest	10,000
Taxable Income (PBT)	50,000

$$\text{Combined leverage} = \frac{C}{PBT} = \frac{1,00,000}{50,000} = 2$$

The combined leverage of '2' indicates that with every increase of ₹ 1 in sales, the taxable income will increase by ₹ 2 (i.e. 1×2). This can be verified by the following computations when the sales increase by 10%

	₹
Sales	2,20,000
Less: variable costs (50%)	1,10,000
Contribution	1,10,000
Less: Fixed operating costs	40,000
EBIT	70,000
Less: Interest	10,000
Taxable Income (PBT)	60,000

It is clear from the above computation that on account of increase in sales by 10%, the profit before tax has increased by 20%.

(b) (i) Computation of Weighted Average Cost of Capital based on existing capital structure

Source of Capital	Existing Capital structure (₹)	Weights (a)	After tax cost of capital (%) (b)	WACC (%) (a) × (b)
Equity share capital (W.N.1)	10,00,000	0.250	10.000	2.500
12% Preference share capital	15,00,000	0.375	12.000	4.500
10% Debentures (W.N.2)	15,00,000	0.375	6.500	2.438
Total	40,00,000	1.000		9.438

Working Notes:

1. Cost of Equity Capital:

$$\begin{aligned}
 K_e &= \frac{\text{Expected dividend}(D_1)}{\text{Current Market Price}(P_0)} + \text{Growth}(g) \\
 &= \frac{₹ 5}{₹ 100} + 0.05 \\
 &= 10\%
 \end{aligned}$$

2. Cost of 10% Debentures

$$\begin{aligned}
 K_d &= \frac{\text{Interest}(1-t)}{\text{Net proceeds}} \\
 &= \frac{₹ 1,50,000 (1-0.35)}{₹ 15,00,000} \\
 &= 0.065 \text{ or } 6.5\%
 \end{aligned}$$

(ii) Computation of Weighted Average Cost of Capital based on new capital structure

Source of Capital	New Capital structure (₹)	Weights (a)	After tax cost of capital (%) (b)	WACC (%) (a) × (b)
Equity share capital (W.N.3)	10,00,000	0.222	12.777	2.836
12% Preference share capital	15,00,000	0.334	12.000	4.000
10% Debentures (W.N.2)	15,00,000	0.333	6.500	2.165

12% Debentures (W.N.4)	5,00,000	0.111	7.800	0.866
Total	45,00,000	1.000		9.867

Working Notes:

3. Cost of Equity Capital:

$$K_e = \frac{\text{₹ 7}}{\text{₹ 90}} + 0.05$$

$$= 12.777\%$$

4. Cost of 12% Debentures

$$K_d = \frac{\text{₹ 60,000 (1-0.35)}}{\text{₹ 5,00,000}}$$

$$= 0.078 \text{ or } 7.8\%$$

(c) Fair Value of Company = Present Value all future cash flows discounted at the expected Rate of return of acquiring company.

WN 1 – Calculation of Cash flows

₹ in Lakhs

YEAR	1	2	3	4	5	6
Contribution (40% on sales)	10	12	14.4	20.16	28.22	35.28
(-) Fixed Cost	-12	-12	-10	-10	-10	-10
NPBT (A)	-2	0	4.4	10.16	18.22	25.28
(-) Losses Set Off	0	0	-2(Setoff)	0	0	0
Taxable Income	0	0	2.4	10.16	18.22	25.28
(-) Tax @ 25% (B)	0	0	0.6	2.54	4.55	6.32
Cash Flow (A – B)	-2	0	3.8	7.62	13.66	18.96
PV OF CASH FLOWS @ 15%	-1.740	0	2.50	4.35	6.79	8.19

Total PV of cash flows (yr 1 to 6) = 20.08 lakhs

(+) PV of cash flow at terminal value (end of Year 6) = $\frac{18.96 + 10\%}{0.15 - 0.10}$

= 417.12 Lakhs

Therefore, PV of above = 417.12 X PV factor (15%, 6th Year)
= 180.20 lakhs

Total fair value of Aryayash limited = 20.08 + 180.20 = 200.28 Lakhs

Note – 1. Discounting rate should be the desired rate of acquiring company i.e. of Vyom Limited

2. Terminal value of cash flows means the cash flows at that point from where it would grow at constant rate. Here it assumed that from 7th year, Cash flows/NPAT will grow at a constant rate and not sales

2. (a) Working Note:

1. Current Liabilities and Current Assets:

Let Current Liabilities be x

$$\text{Given Current ratio} = 2.5$$

$$\text{Current Assets} = 2.5x$$

$$\text{Working Capital} = 2.5x - x = 1.5x$$

$$\text{or } x = 1,20,000 / 1.5 = 80,000$$

$$\text{So Current Liabilities} = 80,000$$

$$\text{And Current Assets} = 80,000 \times 2.5 = 2,00,000$$

2. Closing Stock

$$\text{Given, Quick Ratio} = 1.3$$

$$\frac{\text{Current Assets} - \text{Closing Stock}}{\text{Current Liabilities} - \text{Bank Overdraft}} = 1.3$$

$$\frac{2,00,000 - \text{Closing Stock}}{80,000 - 15,000} = 1.3$$

$$\text{or Closing Stock} = 2,00,000 - 84,500 = 1,15,500$$

$$\text{Opening Stock} = 1,15,000 \times 100 / 110 = 1,05,000$$

3. Debtors

$$\text{Given Debtors Velocity} = 40 \text{ days}$$

$$\frac{\text{Debtors}}{\text{Sales}} \times 365 = 40$$

$$\text{Debtors} = \frac{7,30,000 \times 40}{365} = 80,000$$

4. Gross Profit = 7,30,000 x 10/100 = 73,000

5. Proprietary Fund:

$$\text{Proprietary Ratio} = 0.6$$

$$\frac{\text{Fixed Assets}}{\text{Proprietary Fund}} = 0.6$$

$$\frac{\text{Working Capital}}{\text{Proprietary Fund}} = 0.4$$

$$\text{Proprietary Fund} = \frac{1,20,000}{0.4} = 3,00,000$$

Fixed Assets = 3,00,000 x 0.6 = 1,80,000

Net Profit = 10% of Proprietary Fund = 30,000

M/s Anya Co Ltd.

**Trading and Profit and loss Account for the year ended
31 March 2024**

Particulars	Amount in ₹	Particulars	Amount in ₹
To Opening Stock	1,05,000	By Sales	7,30,000
To Purchase (Balancing Fig.)	6,67,500	By Closing Stock	1,15,500
To Gross Profit	73,000		
	8,45,500		8,45,500
To Operating Expenses (Balancing Figure)	43,000	By Gross Profit	73,000
To Net Profit	30,000		
	73,000		73,000

Balance Sheet as on 31 March 2024

Liabilities	Amount in ₹	Assets	Amount in ₹
Share Capital	2,50,000	Fixed Assets	1,80,000
Reserves & Surplus (Opening bal. + current profit)	50,000		
<i>Current Liabilities</i>		<i>Current Assets</i>	
Bank Overdraft	15,000	Stock	1,15,500
Other Current Liabilities	65,000	Debtors	80,000
		Other Current Assets	4,500
	3,80,000		3,80,000

- (b) **As per Dividend discount model, the price of share is calculated as follows:**

$$P = \frac{D_1}{(1+K_e)^1} + \frac{D_2}{(1+K_e)^2} + \frac{D_3}{(1+K_e)^3} + \frac{D_4}{(1+K_e)^4} + \frac{D_5}{(K_e-g)} \times \frac{1}{(1+K_e)^4}$$

Where,

P = Price per share

K_e = Required rate of return on equity

g = Growth rate

Calculation PV of Dividends

Year	Dividend per share	PVF @ 15%	PV
1	4.4	0.870	3.828
2	4.84	0.756	3.660
3	5.324	0.658	3.503
4	5.856	0.572	3.350
Total			14.341

$$\text{PV of Terminal Value} = \frac{\text{₹}5.856 \times 1.05}{(0.15 - 0.05)^1} \times \frac{1}{(1 + 0.15)^4} = 61.488 \times .572 = 35.171$$

$$\begin{aligned} \text{Intrinsic value of share} &= \text{PV of Dividends} + \text{PV of terminal value} \\ &= 14.341 + 35.171 = \text{₹ } 49.512 \end{aligned}$$

3. 1. In-House Management of Receivables (With Dynamic Discounting)

Particulars:

1. Cash Discount Cost:

- Revised discount rate: 2.5%
- 60% of customers avail discount.
- **Cost of Discount:** ₹ 90,00,000 × 60% × 2.5% = ₹ 1,35,000

2. Bad Debts (Reduced to 0.8% due to dynamic discounting):

- ₹ 90,00,000 × 0.8% = ₹ 72,000

3. Administration Cost: ₹ 1,20,000

4. Cost of Financing Receivables:

- **Working Note 1 (Average Collection Period):** (10 days × 60%) + (60 days × 40%) = 30 days
- **Working Note 2 (Average Receivables):** ₹ 90,00,000 × (30/360) = ₹ 7,50,000
- **Working Note 3 (Cost of Financing):**
 - Cost of Bank Funds: ₹ 7,50,000 × 1/2 × 15% = ₹ 56,250
 - Cost of Owned Funds: ₹ 7,50,000 × 1/2 × 14% = ₹ 52,500
 - **Total Cost of Financing Receivables:** ₹ 1,08,750

Total Cost with In-House Receivables Management and Dynamic Discounting:

Particulars	Amount (₹)
Cash Discount (₹ 90,00,000 × 60% × 2.5%)	1,35,000
Bad Debts (₹ 90,00,000 × 0.8%)	72,000
Admin Cost	1,20,000

Cost of Financing Receivables	1,08,750
Total Cost (In-House with Dynamic Discounting):	4,35,750

2. Factoring Firm's Offer:

Particulars:

1. **Factoring Commission:** ₹ 90,00,000 × 4% = ₹ 3,60,000
2. **Interest Charges on Receivables:** Factor Reserve: 12%, so financing on 88% of receivables. Interest for 25 days: (₹ 90,00,000 - 3,60,000) × 88% × 15% × (25/360) = ₹ 79,200
3. **Cost of Owned Funds (Receivables not factored):** ₹ 13,96,800 × 14% × (25/360) = ₹ 13,580

Owned Funds: (₹ 90,00,000 - 3,60,000) × 12% + 3,60,000 = ₹ 13,96,800

Total Cost with Factoring Firm:

Particulars	Amount (₹)
Factoring Commission (₹ 90,00,000 × 4%)	3,60,000
Interest Charges on Receivables	79,200
Cost of Owned Funds	13,580
Total Cost with Factoring:	4,52,780

3. Impact of Extending Credit Period:

If Zomo Ltd. extends the credit period to 75 days:

- **Sales increase:** 10% of ₹ 120,00,000 = ₹ 12,00,000
New total turnover = ₹ 120,00,000 + ₹ 12,00,000 = ₹ 1,32,00,000
Credit Sales (75%) = ₹ 99,00,000
- **Increased Bad Debts (1.5%):** ₹ 99,00,000 × 1.5% = ₹ 1,48,500
- **Late Payment Penalty:** Customers delaying beyond 60 days (40%):
₹ 99,00,000 × 40% × 5% = ₹ 1,98,000

A. Cash Discount Cost:

- **Discount rate:** 2% (since there's no mention of dynamic discounting in this case)
- **Percentage of customers availing discount:** 60%
- **Calculation:** ₹ 99,00,000 × 60% × 2% = ₹ 1,18,800

B. Bad Debts (Increased to 1.5%):

- **Calculation:** ₹ 99,00,000 × 1.5% = ₹ 1,48,500

C. Administration Costs (Remains the same):

- The administration cost stays fixed at ₹ 1,20,000, as no change in admin structure is mentioned.

D. Cost of Financing Receivables (Based on the new extended credit period):

- **Working Note 1 (Average Collection Period):** Credit period has been extended to 75 days for customers who don't take the discount (40% of customers).
 - **Revised Average Collection Period:** $(10 \text{ days} \times 60\%) + (75 \text{ days} \times 40\%) = 36 \text{ days}$
- **Working Note 2 (Average Receivables):** ₹ 99,00,000 × $(36/360) = ₹ 9,90,000$
- **Working Note 3 (Cost of Financing Receivables):**
 - **Cost of Bank Funds (15%):** ₹ 9,90,000 × $1/2 \times 15\% = ₹ 74,250$
 - **Cost of Owned Funds (14%):** ₹ 9,90,000 × $1/2 \times 14\% = ₹ 69,300$
 - **Total Cost of Financing Receivables:** ₹ 74,250 + ₹ 69,300 = ₹ 1,43,550

Revised Bad Debts after Penalty:

- **Bad debts before penalty:** ₹ 1,48,500
- **Penalty earned:** ₹ 1,98,000
- **Net effect on bad debts:** ₹ 1,48,500 - ₹ 1,98,000 = (-₹ 49,500)
(Zomo Ltd. would effectively earn ₹ 49,500 from penalties, reducing bad debt cost.)

4. Total Cost Calculation:

Now, summing up all the components:

Particulars	Amount (₹)
Cash Discount ($₹ 99,00,000 \times 60\% \times 2\%$)	1,18,800
Net Bad Debts after Penalty ($-₹ 49,500$)	-49,500
Administration Costs	1,20,000
Cost of Financing Receivables	1,43,550
Total Cost (In-House with Extended Credit Period)	₹ 3,32,850

5. Final Decision:

Option	Total Cost (₹)
In-House with Dynamic Discounting	4,35,750
Factoring Firm's Offer	4,52,780
In-House with Extended Credit Period	3,32,850

Recommendation: Zomo Ltd. should **extend the credit period** and continue in-house management. This option will not only reduce costs

(due to lower bad debts offset by penalties) but also increase sales by 10%. Factoring is the least beneficial due to its high commission charges, and dynamic discounting offers only marginal savings compared to the credit extension option.

4. (a) The financing of current assets involves a trade off between risk and return. A firm can choose from short or long term sources of finance. Short term financing is less expensive than long term financing but at the same time, short term financing involves greater risk than long term financing.

Depending on the mix of short term and long term financing, the approach followed by a company may be referred as matching approach, conservative approach and aggressive approach.

In matching approach, long-term finance is used to finance fixed assets and permanent current assets and short term financing to finance temporary or variable current assets. Under the conservative plan, the firm finances its permanent assets and also a part of temporary current assets with long term financing and hence less risk of facing the problem of shortage of funds.

An aggressive policy is said to be followed by the firm when it uses more short term financing than warranted by the matching plan and finances a part of its permanent current assets with short term financing.

- (b) Over-capitalization and its Causes and Consequences

It is a situation where a firm has more capital than it needs or in other words assets are worth less than its issued share capital, and earnings are insufficient to pay dividend and interest.

Causes of Over Capitalization

Over-capitalisation arises due to following reasons:

- (i) Raising more money through issue of shares or debentures than company can employ profitably.
- (ii) Borrowing huge amount at higher rate than rate at which company can earn.
- (iii) Excessive payment for the acquisition of fictitious assets such as goodwill etc.
- (iv) Improper provision for depreciation, replacement of assets and distribution of dividends at a higher rate.
- (v) Wrong estimation of earnings and capitalization.

Consequences of Over-Capitalisation

Over-capitalisation results in the following consequences:

- (i) Considerable reduction in the rate of dividend and interest payments.
- (ii) Reduction in the market price of shares.

- (iii) Resorting to “window dressing”.
 - (iv) Some companies may opt for reorganization. However, sometimes the matter gets worse and the company may go into liquidation.
- (c) “The profit maximisation is not an operationally feasible criterion.” This statement is true because Profit maximisation can be a short-term objective for any organisation and cannot be its sole objective. Profit maximization fails to serve as an operational criterion for maximizing the owner's economic welfare. It fails to provide an operationally feasible measure for ranking alternative courses of action in terms of their economic efficiency. It suffers from the following limitations:
- (i) Vague term: The definition of the term profit is ambiguous. Does it mean short term or long term profit? Does it refer to profit before or after tax? Total profit or profit per share?
 - (ii) Timing of Return: The profit maximization objective does not make distinction between returns received in different time periods. It gives no consideration to the time value of money, and values benefits received today and benefits received after a period as the same.
 - (iii) It ignores the risk factor.
 - (iv) The term maximization is also vague.

OR

- (c) **Modified Internal Rate of Return (MIRR):** There are several limitations attached with the concept of the conventional Internal Rate of Return. The MIRR addresses some of these deficiencies. For example, it eliminates multiple IRR rates; it addresses the reinvestment rate issue and produces results, which are consistent with the Net Present Value method.

Under this method, all cash flows, apart from the initial investment, are brought to the terminal value using an appropriate discount rate (usually the cost of capital). This results in a single stream of cash inflow in the terminal year. The MIRR is obtained by assuming a single outflow in the zeroth year and the terminal cash inflow as mentioned above. The discount rate which equates the present value of the terminal cash inflow to the zeroth year outflow is called the MIRR.